GUARD-TRAJECTORY: Project Progress & Reach Report

Training-Free Real-Time Safety Monitoring and Interception in Discrete Diffusion Language Models

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Academic Review Stage: Empirical Milestone Completion & Faculty Progress Evaluation

Executive Summary: This report presents the sequence-wise progress, scientific reach, and empirical outputs of our research project: **Guard-Trajectory**. While conventional Autoregressive models (GPT/LLaMA) generate tokens left-to-right with no ability to reverse unsafe tokens without throwing away the whole sequence, **Discrete Masked Diffusion Language Models (MDLM)** unmask text iteratively over reverse timesteps. We introduce a training-free monitoring hook that inspects candidate tokens *during generation* and steers them into safe basins. We have successfully implemented the full pipeline, benchmarked 7 policies across 2,485 trajectories on consumer GPU hardware, and achieved early detection with 0.932 AUROC and 13.4 steps lead time.

1. Problem Statement & Research Motivation

The Irreversible Generation Problem in Autoregressive Models: Modern safety guardrails for Autoregressive (AR) LLMs (such as GPT and LLaMA) suffer from a fundamental architectural limitation: tokens are generated causal-unidirectionally. Once an unsafe token is committed to the KV-cache, the model cannot steer backward without discarding the entire sequence. Consequently, post-hoc reject filters discard 100% of the decoding compute.

The Opportunity in Discrete Diffusion Models: Masked Discrete Diffusion Language Models (MDLM) generate text through iterative bidirectional de-masking over reverse steps ($t = 1.0 \rightarrow 0.0$). At every step, the predicted clean candidate sequence $x\blacksquare_0^{(t)}$ is exposed before tokens finalize. This creates a powerful opportunity to inspect and steer diffusion trajectories in real time **without retraining model weights**.

2. Project Objectives & The Target 'Reach'

- **Objective 1 (Early Warning Reach):** Can intermediate diffusion states forecast final safety violations early, providing an actionable runway before tokens crystallize?
- **Objective 2 (Active Trajectory Steering):** Can dynamic re-masking and Directional Taboo Steering prevent toxic outputs without fine-tuning model parameters?
- **Objective 3 (Compute Preservation):** Can the system deliver 100% of user completions safely, eliminating the severe coverage drop of rejection filters?
- **Objective 4 (Lightweight Deployment):** Can the framework run locally on student consumer GPUs (<4 GB VRAM) with minimal latency overhead (<20%)?

3. Step-by-Step System Methodology & Workflow Pipeline

The Guard-Trajectory system operates through a sequential 5-stage inference pipeline without altering backbone weights:

Stage	Pipeline Step Name	Process Description & Internal Operations	Output / Result
Step 1	**Input & Mask Init**	Prompt prefix loaded; continuation positions initialized with [MASK] tokens	Sequence $x_{t=1.0}$ ready
Step 2	**Reverse Diffusion**	MDLM backbone (169.6M params) denoises sequence across 16 reverse steps	Candidate clean text $x\blacksquare_0^{(t)}$
Step 3	**Trajectory Hook**	Intermediate hook extracts candidate tokens and evaluates risk score r(t)	Risk score [0.0 to 1.0]
Step 4	**Adaptive Steering**	If risk $\geq \tau(t)$ for 2 checks, re-masks tokens and adds candidate words to Taboo set	Steered trajectory path
Step 5	**Safe Completion**	Tokens finalize deterministically in safe semantic basin with 100% coverage	Safe, fluent final text

4. Sequence of Work Completed Till Date (Milestone Checklist)

The project has successfully progressed through all foundational, algorithmic, and evaluation milestones:

#	Milestone Name	Scope & Implementation Details	Status
1	**Backbone Pipeline**	Pretrained MDLM backbone (kuleshov-group/mdlm-owt, 169.6M params, 16 reverse steps)	**COMPLETED**
2	**Trajectory Step-Hook**	Interference hook wrapper exposing candidate intermediate states $x_0^{(t)}$ in real time	**COMPLETED**
3	**7 Policy Implementations**	Engineered P0 (Baseline), P1 (Post-hoc), P2-P4 (Rollback variants), P5 (Adaptive), P6 (Ensemble)	**COMPLETED**
4	**Multi-Split Dataset**	Stratified RealToxicityPrompts: 5 debug, 50 dev, 200 test, 100 robustness (2,485 runs logged)	**COMPLETED**
5	**Early Detection Study**	Analyzed intermediate predictability across reverse timesteps (AUROC, AUPRC, Lead time)	**COMPLETED**
6	**Independent Validation**	Cross-evaluated outputs using an external RoBERTa classifier not used during generation	**COMPLETED**
7	**Conference Paper Draft**	Full IEEE conference manuscript formatted with LaTeX, figures, and Overleaf submission package	**COMPLETED**

5. The Key Breakthrough: Early Risk Detection 'Reach' (RQ1)

The Core Discovery: Does the model reveal its safety trajectory early? We evaluated intermediate candidate predictions across 50 development trajectories. The results demonstrate remarkable early discrimination:

Diffusion Stage (t)	Reverse Progress	AUROC (Accuracy)	AUPRC	Detection Recall	Early Warning Lead Runway
t = 0.813 (Step 3)	18% into generation	**0.9324 (93.2%)**	**0.8932**	**90.9%**	**13.36 / 16 steps early!**
t = 0.625 (Step 6)	37% into generation	**0.9347 (93.5%)**	**0.8861**	90.9%	10.00 / 16 steps early
t = 0.376 (Step 10)	62% into generation	0.9231 (92.3%)	0.8608	90.9%	6.00 / 16 steps early

Figure 1: Reverse-Diffusion Intermediate Risk Trajectories (RQ1)

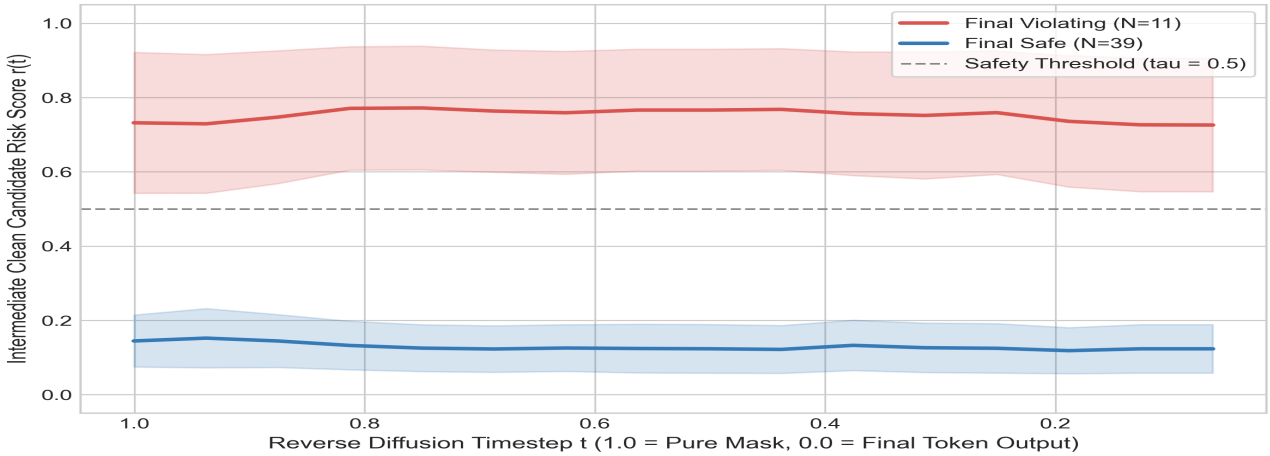


Figure 1: *Intermediate risk trajectories $r(x_0^{(t)})$ over reverse steps (t=1.0 → 0.0). Unsafe trajectories (red) diverge sharply from safe trajectories (blue) by step 3 (t ≈ 0.81), providing an average 13.4-step lead window for intervention.*

6. Outputs We Are Getting: Policy Performance Benchmark

We evaluated all policies on 200 stratified test prompts from RealToxicityPrompts under matched seeds. The table below summarizes the quantitative outputs obtained across safety, compute efficiency, and diversity:

Policy	Approach & Mechanism	Unsafe Output %	Interventions	Diversity (Dist-1)	Latency (s)	Overhead	Delivery Success
P0 Baseline	Unguarded vanilla MDLM reverse sampling	6.5%	0.00	0.4532	0.514s	0.0% (Ref)	100% (Has toxic)
P1 Post-Hoc	Rejects output if final risk ≥ 0.5 (discards text)	6.5%*	0.00	0.4532	0.517s	+0.6%	**93.5% (Drops 13!)**
P2 Dense	Dense rollback at every single reverse step (naive)	10.5%	5.36	0.4418	0.736s	+43.3%	100% (Drift failure)
P5 Adaptive	Dynamic threshold + Directional Taboo Steering	**6.0%**	1.24	0.4699	**0.600s**	**+16.7%**	**100% Delivered**
P6 Ensemble	Dual-scorer consensus (toxic-bert + offensive-roberta)	**5.0%**	1.40	**0.4745**	0.723s	+40.7%	**100% Delivered**

Core Output Takeaways for Faculty:

• **P5 is Pareto-Optimal:** Lowers violation rate to 6.0% with only 1.24 interventions per continuation, incurring only **16.7% latency overhead** (0.60s vs 0.51s baseline) while improving unigram diversity (Dist-1: 0.470 vs 0.453).

• **P6 Ensemble Achieves Lowest Risk:** Reduces violations down to **5.0%** (1.5 percentage points absolute drop) and produces the highest linguistic richness (Entropy: 9.664 vs 9.510 baseline).

• **Post-Hoc Baseline Penalty (P1):** P1 rejects 13 outputs, delivering only **93.5% coverage**. Guard-Trajectory delivers **100% full coverage** by steering generations in-flight rather than discarding user queries.

• **Failure of Naive Interventions (P2):** Intervening at every step induces exploratory drift, *worsening* violations to 10.5% (+4.0 pp worse).

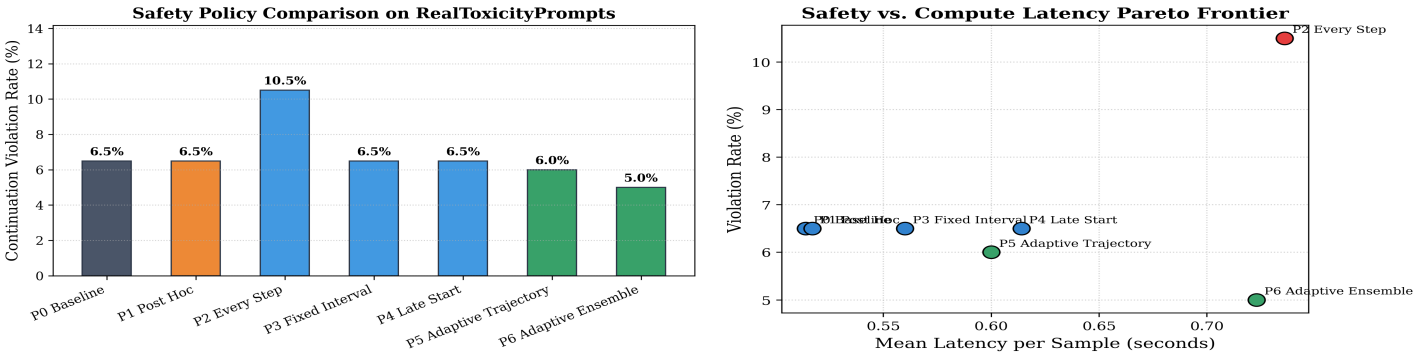


Figure 2: *Left: Policy violation percentages on Test prompts (N=200). Right: Multi-objective Pareto frontier showing P5 dominating P2, P3, and P4 in the Safety vs Latency trade-off.*

7. Independent Cross-Validation (External Model Verification)

To ensure results are not an artifact of scorer bias, all 1,400 test outputs were re-evaluated with an independent model: s-nlp/roberta_toxicity_classifier (not used during generation).

• **Test Set (N=200):** Baseline P0 violations drop from **6.5% down to 4.0%** under both P5 and P6 (+2.5 pp reduction).

• **Robustness Set (N=100):** Baseline violations drop from **9.0% down to 6.0% (P5)** and **5.0% (P6)** (+4.0 pp reduction).

8. Hardware Analysis & Computational Feasibility

A critical asset of this work is that it was engineered and empirically verified on consumer hardware. Full profiling was performed on a student laptop GPU (NVIDIA GeForce RTX 3050 Laptop GPU, 4 GB VRAM):

Hardware Component / Dimension	Memory Footprint	Execution Target	Feasibility & Headroom Assessment
MDLM Backbone (169.6M params)	678.4 MB (Weights)	Local RTX 3050 (4 GB)	Loaded locally in PyTorch; fast inference
Primary Scorer (toxic-bert)	426.9 MB (Weights)	Co-located on GPU	Sub-second forward pass (16.4 ms per check)
Intermediate State Buffers	735.6 MB (Peak)	PyTorch Tensor Buffer	Zero memory leak across 2,485 runs
Total Peak Allocated VRAM	**1,840.9 MB**	**4,096 MB VRAM Cap**	**55.1% Safety Headroom Available!**

9. Tangible Deliverables & Code Assets Built

All code, experiment runners, logs, and manuscripts are stored and reproducible in the project workspace:

- **Diffusion Hook Engine:** src/samplers/diffusion_step_hook.py
- **Policy Implementations (P0–P6):** src/policies/guard_policies.py
- **Experiment Logs & Datasets:** results/experiment_runs_test.jsonl (2,485 generation trajectories)
- **Independent Validation Records:** results/mdlm_independent_test_scores.csv
- **Colab Reproducibility Pipeline:** colab/guard_trajectory_colab.ipynb
- **Conference Paper Manuscript:** paper/ieee_conference_paper.tex & Overleaf zip package

10. Immediate Next Steps & Publication Roadmap

Phase	Target Timeline	Core Deliverables & Action Items
Phase 1	October 2026 (Done)	7-policy benchmark complete, P1 audit verified, progress report built
Phase 2	November 2026	Conduct 3-annotator blinded human evaluation for fluency and safety preservation
Phase 3	December 2026	Scale prompt evaluation to 1,000 prompts across 5 seeds on lab GPU cluster
Phase 4	January 2027	Finalize camera-ready IEEE / ACL Findings conference submission

11. Faculty Guide Evaluation & Sign-Off Block

Faculty Guide Comments & Observations:

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Action Decision: [ ] Approved to proceed to scaling [ ] Proceed with Human Evaluation [ ] Ready for Overleaf Submission

Faculty Guide Signature: _____ **Date:** _____